
Instructions. (15 points) Sections 14.4–14.5. Show relevant work for full credit.

(8^{pts}) 1. A grain silo consists of a cylinder of radius $r = 3$ m and height $h = 8$ m topped by a hemisphere of radius $r = 3$ m. The silo is built from wooden planks that are 0.05 m thick.

(a) (4 pts) The volume of the silo is $V = \pi r^2 h + \frac{2}{3}\pi r^3$. Find the total differential dV .

Solution:

$$dV = \frac{\partial V}{\partial r} dr + \frac{\partial V}{\partial h} dh = (2\pi r h + 2\pi r^2) dr + \pi r^2 dh$$

(b) (4 pts) Determine the appropriate values of dr and dh for this problem, then use differentials to estimate the total volume of wood in the silo walls, roof, and floor.

Solution:

The planks add one layer outward on the cylindrical walls and the dome, so $dr = 0.05$.

The silo has a flat wooden floor (bottom) but **no flat top** — the hemisphere replaces the lid. So only the floor adds to the height: $dh = 0.05$.

$$\begin{aligned} dV &= (2\pi(3)(8) + 2\pi(9))(0.05) + \pi(9)(0.05) \\ &= (48\pi + 18\pi)(0.05) + 0.45\pi \\ &= 3.3\pi + 0.45\pi \\ &= \boxed{3.75\pi \approx 11.78 \text{ m}^3} \end{aligned}$$

(7^{pts}) **2.** Let $w = xy + yz + xz$, where $x = s^2 + t$, $y = st$, and $z = t^2$.

(a) (3 pts) Draw the tree diagram and write the chain rule expression for $\frac{\partial w}{\partial t}$.

Solution:

Tree diagram: w depends on x, y, z , each of which depends on s and t .

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial t}$$

(b) (4 pts) Evaluate $\frac{\partial w}{\partial t}$ at $(s, t) = (1, 2)$.

Solution:

At $(s, t) = (1, 2)$: $x = 1 + 2 = 3$, $y = 2$, $z = 4$.

Partials of w :

$$\frac{\partial w}{\partial x} = y + z = 6, \quad \frac{\partial w}{\partial y} = x + z = 7, \quad \frac{\partial w}{\partial z} = x + y = 5$$

Partials of x, y, z with respect to t :

$$\frac{\partial x}{\partial t} = 1, \quad \frac{\partial y}{\partial t} = s = 1, \quad \frac{\partial z}{\partial t} = 2t = 4$$

Therefore:

$$\frac{\partial w}{\partial t} = 6(1) + 7(1) + 5(4) = 6 + 7 + 20 = \boxed{33}$$